Republic of the Philippines
CAVITE STATE UNIVERSITY
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COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY
Department of Information Technology

Name: _____ Student No.: _____
Course / Yr. / Sec.: _____ Date: _____

Portfolio #10: Discussion on Physical Security Challenges

Objectives

- To evaluate the effectiveness of physical security measures, facility controls, and environmental safety systems in protecting critical data centers.
- To analyze the structural risks associated with Bring Your Own Device (BYOD) environments and formulate Mobile Device Management (MDM) solutions.
- To design an endpoint hardening strategy and differentiate between legacy EPP and modern EDR systems.
- To collaborate as a cybersecurity consulting team to resolve complex physical and endpoint vulnerabilities in a corporate scenario.

Background / Scenario

The Case of Aegis Healthcare Processing (Groupwork Scenario)

Aegis Healthcare Processing recently expanded its operations into a newly constructed, three-story facility. Because they handle highly sensitive electronic health records, their network perimeter is heavily fortified with Next-Generation Firewalls and Intrusion Prevention Systems. However, their physical and endpoint security posture has been severely neglected.

Last month, a series of concerning incidents occurred:

1. **The Tailgating Incident:** An unauthorized individual, dressed as a delivery driver, followed a legitimate employee through the main server room doors without scanning a badge.
2. **The Power Anomaly:** A prolonged brownout caused voltage drops that permanently damaged several critical database servers because the facility was only using basic power strips.
3. **The BYOD Data Leak:** A doctor's personal smartphone, which contained downloaded patient records, was stolen at a coffee shop. The hospital had no way to wipe the corporate data without also erasing the doctor's personal photos, leading to a legal and privacy dilemma.
4. **The HVAC Dust Issue:** Construction near the facility caused heavy dust accumulation inside the server racks because the server room doors pull air *inward* when opened.

The Chief Information Security Officer (CISO) has hired your group as an external Cybersecurity Consulting Team. You must convene to discuss these physical security challenges and draft a comprehensive, collaborative **Physical and Endpoint Security Strategy Report**. This report will outline how Aegis Healthcare must overhaul its facility controls, environmental safety, and mobile device policies.

Instructions

Part 1: Facility Controls & Environmental Safety
Collaborate with your team to address the physical vulnerabilities of the Aegis facility.

- **Biometrics and Mantraps:** To prevent another "tailgating" or "piggybacking" incident, propose the installation of a Mantrap (Security Vestibule) paired with a biometric scanner. Explain the mathematics of biometric accuracy to the board—specifically defining False Accept Rate (FAR), False Reject Rate (FRR), and why a lower Equal Error Rate (EER) is critical for their server room.
- **Fire & HVAC Physics:** The current server room uses water sprinklers and standard air conditioning. Explain why water (Class A) suppression is dangerous for electronics, and recommend a modern "Clean Agent" (e.g., FM-200/Inergen). Furthermore, explain how applying "Positive Pressure" via the HVAC system will solve their dust accumulation problem.
- **Power Anomalies:** Differentiate between a surge, sag, brownout, and blackout. Recommend a specific Uninterruptible Power Supply (UPS) architecture (e.g., double-conversion online UPS) to protect against the "dirty power" that fried their servers.

Part 2: Endpoint Protection & Mobile Security (BYOD)
Address the vulnerabilities of the distributed workforce and stolen devices.

- **The BYOD Dilemma:** Explain the structural risk of allowing employees to use personal phones for healthcare data. Propose a Mobile Device Management (MDM) solution that utilizes "Containerization" (Sandboxing). Explain how this specifically solves the dilemma of performing a "Remote Wipe" on a stolen phone without destroying the employee's personal files.
- **EPP vs. EDR:** The current workstations use legacy Endpoint Protection Platforms (Traditional Antivirus). Argue why the company must upgrade to Endpoint Detection and Response (EDR). Focus your argument on how EDR monitors *behavior* to stop modern "Fileless Malware," whereas traditional AV relies only on signatures.

Part 3: Hybrid Cryptography & SSL/TLS Implementation
Design a baseline hardening protocol for all new Aegis workstations.

- **Hardening Steps:** Outline a strict 4-step hardening process for the facility's desktop computers (must include disabling unnecessary services, closing ports, least privilege/credential management, and local logging).
- **Data Interception:** The CISO is worried about physical interception. Briefly explain the threat of Hardware Keyloggers and TEMPEST (Spurious Emissions), and state why traditional software firewalls cannot stop these specific attacks.

Reflection

Discuss these questions as a group and synthesize a unified response)

1. Why is physical security universally considered the "Outer Shell" of the CIA Triad? How does the failure of a physical lock or a power supply render the most expensive Next-Generation Firewall completely useless?
2. How did working in a 5-person team help your group evaluate these physical security challenges more comprehensively than working alone?

Submission Format

- **File Naming Convention:** Portfolio10_<LeaderSurname> (e.g., Portfolio10_DelaCruz)
- **Format:** Save and submit as a PDF file.
- **Platform:** Upload to the designated Google Classroom submission bin.
- **Group Requirement:** Group yourselves into teams with a minimum of five members, and ensure that all members' names are listed at the top of the document. Coordinate with your class representative for your official grouping. Only the Leader needs to upload the final file on behalf of the group. Remove all headers and rubric from your submission.

Rubric

Criterion	Excellent (25 Points)	Good (16–20 Points)	Needs Improvement (11–15 Points)	Poor (0–10 Points)	Points
Following Instructions & Formatting	All parts complete, cleanly formatted as a PDF, correct file name, and all 5 group member names are prominently listed.	All major parts complete, minor omission or slight deviation in file naming or formatting.	Significant portion incomplete or poorly formatted; missing group members' names.	Not submitted or completely disregards format and instructions.	/25
Physical & Environmental Security Analysis	Flawlessly explains FAR/FRR/EER, positive pressure, clean agents, and UPS types. Deeply tied to the Aegis scenario.	Correctly explains the physical controls; justification relies on basic definitions with minimal scenario application.	Confuses basic physical security functions (e.g., FAR vs FRR) or fails to address the specific Aegis problems (dust, brownouts).	Core physical security concepts are fundamentally wrong or entirely missing.	/25
Endpoint & Mobile Defenses Strategy	Highly compelling explanation of MDM containerization and EDR behavioral monitoring. Hardening steps are exact and logical.	Good strategy provided; correctly identifies MDM and EDR strengths but lacks depth in explaining the BYOD remote wipe dilemma.	Argument is thin; fails to properly distinguish between traditional Antivirus (EPP) and EDR, or misunderstands BYOD risks.	No clear strategy presented, or reasoning is entirely illogical.	/25
AI Originality & Group Effort	Clear original thought, student's own words, profound group reflection, excellent synthesis of physical/digital overlaps, no AI boilerplate.	Students' own work, reflection is somewhat generic but adequate synthesis of concepts.	Vague, overly formal, generic phrasing that raises AI concerns; heavily relies on standard definitions without group synthesis.	Clearly AI-generated with little modification, or plagiarized content.	/25
TOTAL SCORE					/100